THE REVERSE-ENGINEERED MASTER PROTOCOL

Electrical Instrumentation Part-I & II: Ultimate Roadmap

Integrating the 4-Year Syllabus, Capacitive Discs, and Vertical Diagram Typography

Academic Masterclass

Time Remaining: 48 Hours

Contents

The 4-Year Topic Matrix (What BJB Sir Actually Tests)

Based on the rigorous analysis of the 2022, 2023, 2024, and 2025 examination papers, the syllabus strictly revolves around these core engineering blocks. You must command these to pass.

The Historical Question Bank Checklist

1. **Capacitive Transducers:**
   - Variable Air Gap (Hyperbolic Non-Linearity)
   - **The Missing Link:** Capacitive Disc (Angular/Overlapping Area)
   - Push-Pull Differential configuration (Linearization Proof)
   - Solid Dielectric Insertion (Thickness Gauge)
   - Capacitive Humidity Sensors & Liquid Level Gauges
2. **Inductive Transducers (LVDT):**
   - Exact Architecture (Longitudinal Core Slot)
   - The Residual Null Voltage Crisis (Harmonics & Quadrature)
   - Phase Shift Compensation (R-C Low Pass Filter)
   - Phase-Sensitive Demodulation (PSD) vs. Ordinary Rectifiers
3. **Piezoelectric Transducers:**
   - The d and g Constants Relation ($g = d/\epsilon$)
   - Why Steady State (Static) Gain is mathematically zero
   - The mandatory Charge Amplifier (Virtual Grounding C_{cable})
   - Seismic Accelerometer Transfer Function (2^{nd} order mech $\times$ 1^{st} order elec)
4. **Interfacing & "Other Topics" (Part II Elements):**
   - Transformer Ratio Bridges (Immunity to Stray Capacitance)
   - Ultrasonic Transit-Time Flowmeters (Wetted vs Non-Wetted)
   - Logarithmic Amplifiers for NTC Thermistors (Linearizing $1/T$)
   - Filters (Butterworth vs Chebyshev properties)

Phase 1: The Capacitive Disc (Angular) Transducer

This is the exact problem highlighted in past exams requiring proof of sensitivity proportionality. It utilizes rotational movement rather than linear translation.

Grassroots Analytical Derivation

The transducer consists of two parallel semi-circular discs (plates). 1. The **Stator** is rigidly fixed. 2. The **Rotor** is attached to the measurement shaft. Both plates have a maximum effective radius r. They are separated by a constant, rigid air gap d.

When the shaft rotates by an angle θ (measured strictly in radians), the plates overlap. The mathematical area of a sector of a circle is $A = \frac{1}{2}r^2\theta$. Substituting this overlapping area into the fundamental parallel-plate capacitance equation ($C = \frac{\epsilon A}{d}$):

$$C(\theta) = \frac{\epsilon\left(\frac{1}{2}r^2\theta\right)}{d} = \frac{\epsilon \cdot r^2 \cdot \theta}{2d} \tag{1}$$

Defining the Sensitivity (S): Sensitivity is the rate of change of capacitance with respect to the input angle θ.

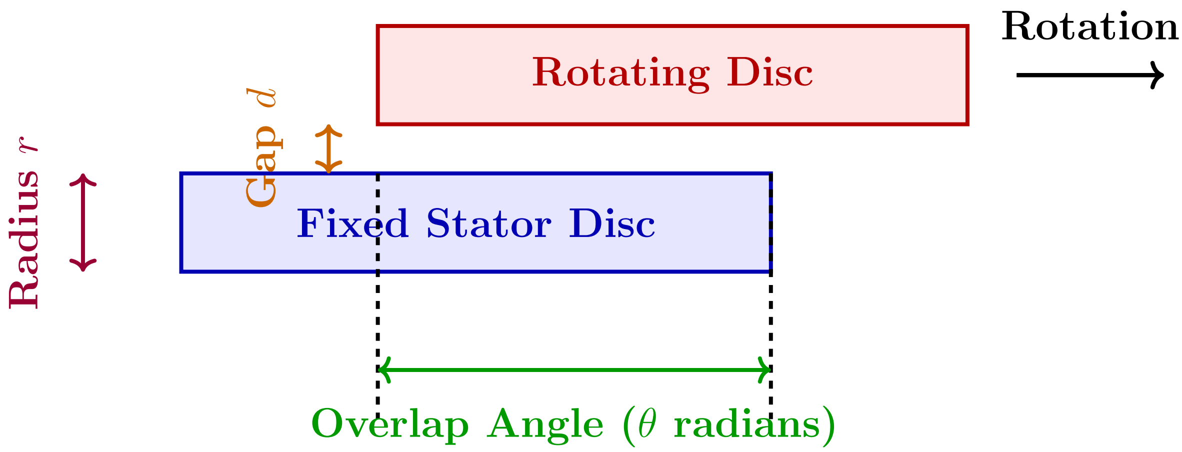
$$S = \frac{dC}{d\theta} = \frac{d}{d\theta}\left[\frac{\epsilon \cdot r^2 \cdot \theta}{2d}\right] = \frac{\epsilon \cdot r^2}{2d} \tag{2}$$

BJB Sir's Required Conclusions: Looking purely at the final mathematical block for S:

- θ is absent from the equation. Therefore, Sensitivity is a **constant**, proving the device is perfectly **Linear**.
- Sensitivity is **directly proportional to the square of the radius** (r^2).
- Sensitivity is **inversely proportional to the gap distance** (d).

Orthogonal Diagram with Vertical Typography

To absolutely prevent horizontal text overlapping, all side annotations are rigorously rotated 90 degrees. The angular geometry is represented as an unwrapped orthogonal block for maximum clarity.


Rotation
Rotating Disc
Gap d
Radius r
Fixed Stator Disc
Overlap Angle (θ radians)

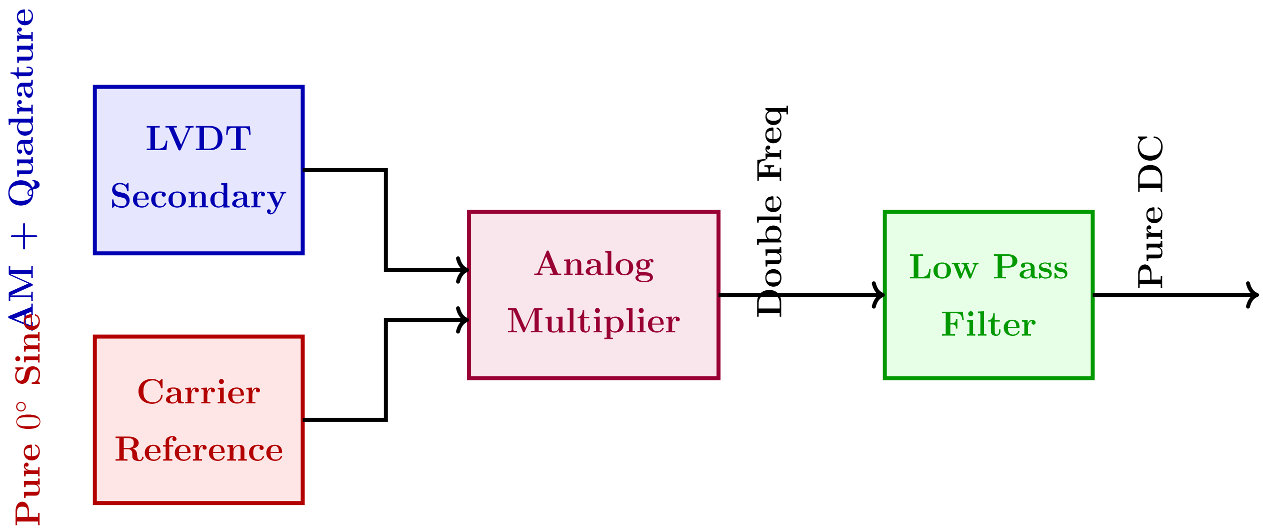
Phase 2: Mastering LVDT & Inductive Systems

The Null Voltage Crisis & Resolution

If BJB Sir asks why $E_{out} \neq 0$ at the exact physical center of the LVDT, you must provide this two-part grassroots explanation:

1. Harmonic Distortion: The nickel-iron core saturates, creating odd-harmonics in the magnetic flux. These harmonics do not possess the same spatial wavelengths as the fundamental frequency and therefore fail to cancel in series opposition. **2. Quadrature Stray Capacitance:** Leakage currents flow through stray parasitic capacitors between the primary and secondary coils. Current through a capacitor ($i = C\frac{dv}{dt}$) produces a 90° phase shift. This quadrature voltage physically cannot cancel against the fundamental 0° induced magnetic voltage.

The Solution Diagram: Phase Sensitive Demodulation (PSD) (Vertical text utilized for dense labeling)



Mathematical Proof: $V_{mult} = [E_{error}\sin(\omega_c t + 90°)] \times [V_{ref}\sin(\omega_c t)]$. This yields $\frac{E_{error}V_{ref}}{2}\sin(2\omega_c t)$. The Low Pass Filter integrates this to exactly **0 Volts DC**. Quadrature error is destroyed.

Phase 3: The Piezoelectric Charge Amplifier

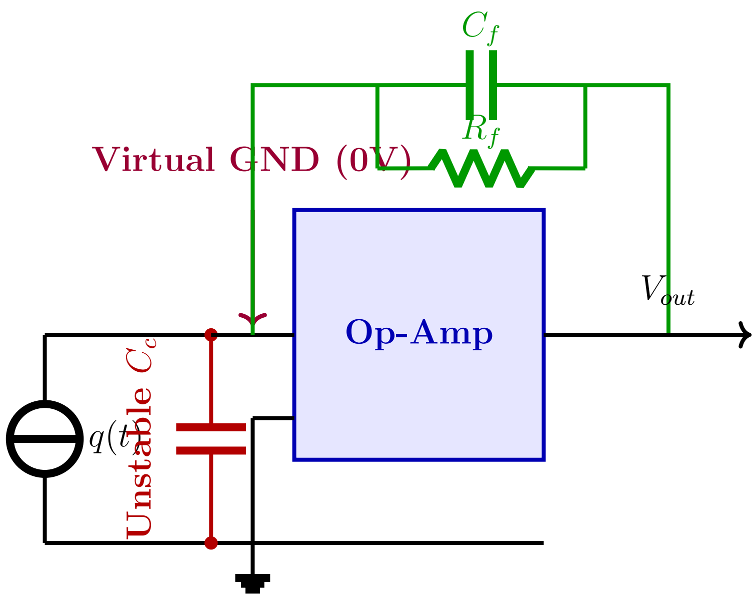
BJB Sir despises voltage amplifiers for piezoelectric crystals. You must be able to justify the Charge Amplifier using the Virtual Ground concept.

The Cable Capacitance Catastrophe

If a crystal (Capacitance C_p) is connected to a standard amplifier via a coaxial cable (Capacitance C_c), the measured voltage is $V = \frac{q}{C_p+C_c+C_a}$. If the cable vibrates, C_c fluctuates wildly. Because C_c is in the denominator, the voltage violently spikes, burying the true vibration signal in noise.

The Charge Amplifier Solution

By placing a capacitor (C_f) in the negative feedback loop of an Op-Amp, the inverting terminal is forcefully clamped to 0V (Virtual Ground). **If the node is at 0V, absolutely no voltage can develop across the cable capacitance** C_c. Therefore, the cable stores zero charge ($Q = C \times 0 = 0$). All generated charge q is forcefully swept directly into C_f, yielding a perfectly stable output: $V_{out} = -q/C_f$.



Phase 4: The "Other" Core Topics (Part II)

4.1 Ultrasonic Transit-Time Flowmeter

The Mathematics: Two sensors shoot pulses diagonally across a flowing liquid. The downstream pulse travels faster ($c + v\cos\theta$) than the upstream pulse ($c - v\cos\theta$). The differential transit time Δt is calculated as:

$$\Delta t = \frac{L}{c - v\cos\theta} - \frac{L}{c + v\cos\theta} \approx \frac{2Lv\cos\theta}{c^2} \tag{3}$$

Temperature Dependency Trap: Because the speed of sound c is fiercely dependent on temperature, Δt will fluctuate with temperature even if fluid velocity v is constant.

4.2 Logarithmic Amplifiers for NTC Thermistors

The Flaw: NTC Thermistors have an exponential resistance curve: $R_T = A\exp(\beta/T)$. **The Log Amp Fix:** By feeding the thermistor voltage ($V_{in} = I_c R_T$) into a Logarithmic Amplifier ($V_{out} = K\ln(V_{in})$), the exponential term is mathematically crushed into a linear slope.

$$V_{out} = K\ln\left[I_c A\exp\left(\frac{\beta}{T}\right)\right] = K\ln(I_c A) + K\beta\left(\frac{1}{T}\right) \tag{4}$$

Conclusion: The output is strictly linearized with respect to the **Inverse Temperature** ($1/T$).

4.3 Transformer Ratio Bridge (Capacitance)

Why use it? Standard bridges fail at measuring pF capacitors because stray grounding capacitance ruins the node voltages. **The Fix:** A tightly coupled transformer forces the voltage ratio V_1/V_2 to absolutely equal the physical turns ratio N_1/N_2. Because the transformer possesses near-zero internal impedance, it aggressively holds these voltages steady, completely ignoring and overpowering any parasitic stray capacitance attempting to drain the node to ground.